

Lecture 5/Module 1

DC Circuit Analysis and Theorems

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Mode of Evaluation:



Mode	Syllabus	Exam Time
CAT I	Module 1 (CO 1)	August 09-16
Quiz 1	Module 3 (CO 3)	Early Sept
CAT II	Module 2,3 (CO 2,3)	Sept 27- Oct 04
Quiz 2	Module 5 (CO 5)	Mid Oct
Course Project	Module 4 (CO 4)	03 November 2026, Last Instruction Day of Semester
FAT	All Modules	Nov-Dec

CO: Course Outcome (please refer to the Lecture 01)

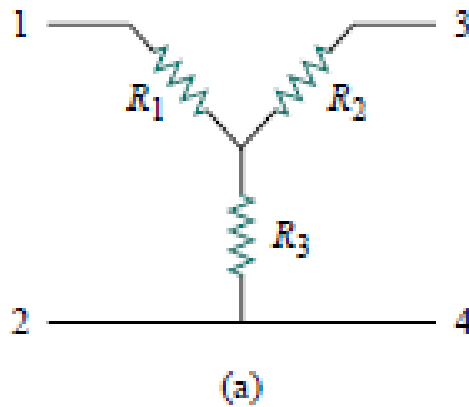
Lecture Topics



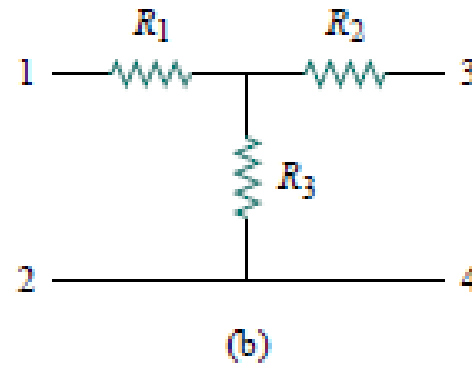
- Wye-Delta Transformation
- Nodal Analysis, Super Node
- Mesh Analysis, Super Mesh

Wye-Delta Transformations

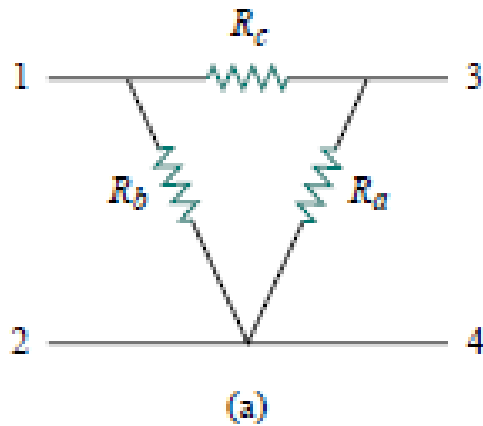
Y



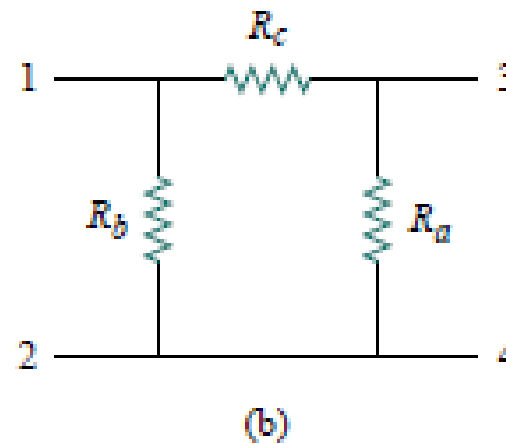
T



Δ



π



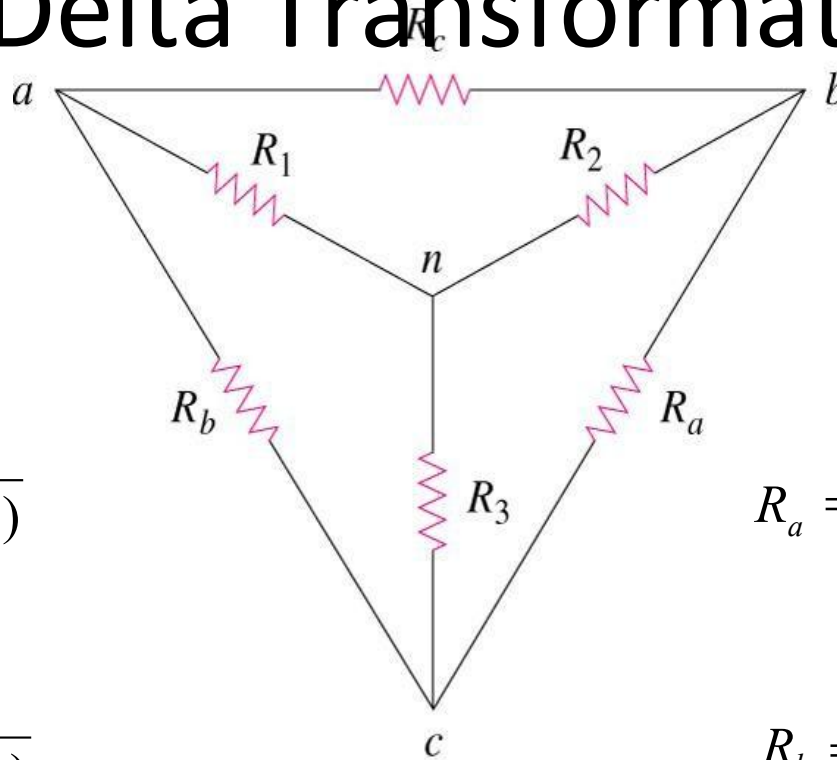
Wye-Delta Transformations

Delta -> Star

$$R_1 = \frac{R_b R_c}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{R_c R_a}{(R_a + R_b + R_c)}$$

$$R_3 = \frac{R_a R_b}{(R_a + R_b + R_c)}$$



Star -> Delta

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

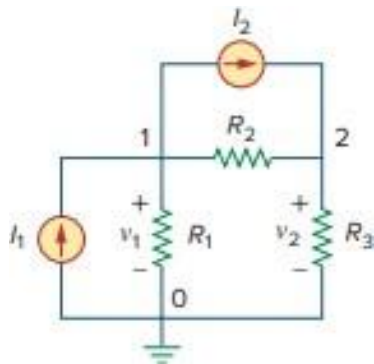
$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

Nodal analysis

In nodal analysis, we are interested in finding the node voltages.

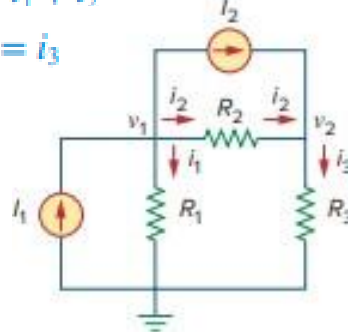
Steps to determine Node Voltages:

1. Select a node as the reference node.
2. Assign voltages $v_1, v_2, \dots, v_n - 1$ to the remaining $n - 1$ nodes. The voltages are referenced with respect to the reference node.
3. Apply KCL to each of the $n - 1$ not referenced nodes. Use Ohm's law to express the branch currents in terms of node voltages.
4. Solve the resulting simultaneous equations to obtain the unknown node voltages.



$$I_1 = I_2 + i_1 + i_3$$

$$I_2 + i_2 = i_3$$



$$\begin{aligned} i_1 &= \frac{v_1 - 0}{R_1} & \text{or} & & i_1 &= G_1 v_1 \\ i_2 &= \frac{v_1 - v_2}{R_2} & \text{or} & & i_2 &= G_2 (v_1 - v_2) \\ i_3 &= \frac{v_2 - 0}{R_3} & \text{or} & & i_3 &= G_3 v_2 \end{aligned}$$

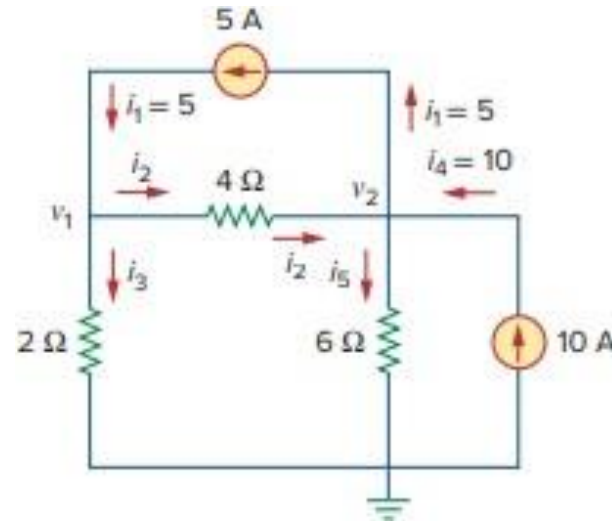
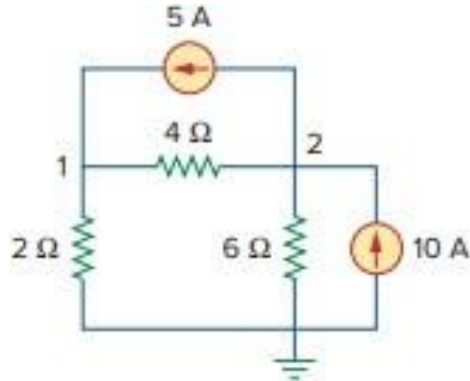
$$\begin{aligned} I_1 &= I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2} \\ I_2 + \frac{v_1 - v_2}{R_2} &= \frac{v_2}{R_3} \end{aligned}$$

$$I_1 = I_2 + G_1 v_1 + G_2 (v_1 - v_2)$$

$$I_2 + G_2 (v_1 - v_2) = G_3 v_2$$

Nodal Analysis

Calculate the node voltages in the circuit



At node 1,

$$i_1 = i_2 + i_3 \quad \Rightarrow \quad 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$

$$20 = v_1 - v_2 + 2v_1$$

At node 2,

$$3v_1 - v_2 = 20 \quad \text{-----1}$$

$$i_2 + i_4 = i_1 + i_5 \quad \Rightarrow \quad \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6}$$

$$3v_1 - 3v_2 + 120 = 60 + 2v_2$$

$$-3v_1 + 5v_2 = 60 \quad \text{-----2}$$

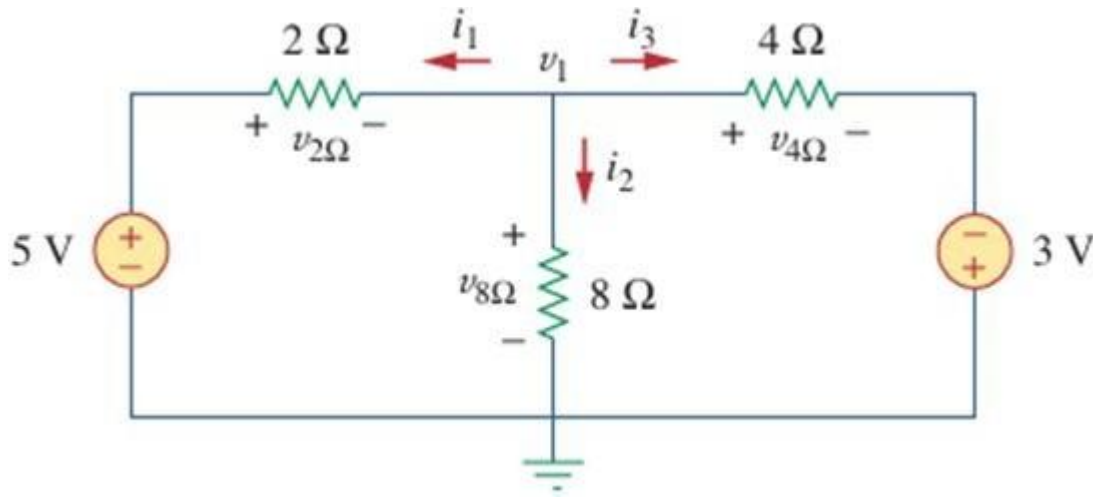
Solve the 2 simultaneous equations:

$$v_1 = 13.33\ \text{V}$$

$$v_2 = 20\ \text{V}$$

Solve for $i_{8\Omega}$ using Nodal analysis method.

Nodal Analysis



$$i_{8\Omega} = i_2, \quad i_2 = \frac{v_1}{8}, \quad i_{8\Omega} = \frac{v_1}{8}$$

$$\frac{v_1 - 5}{2} + \frac{v_1 - 0}{8} + \frac{v_1 + 3}{4} = 0$$

Now we can solve for v_1 .

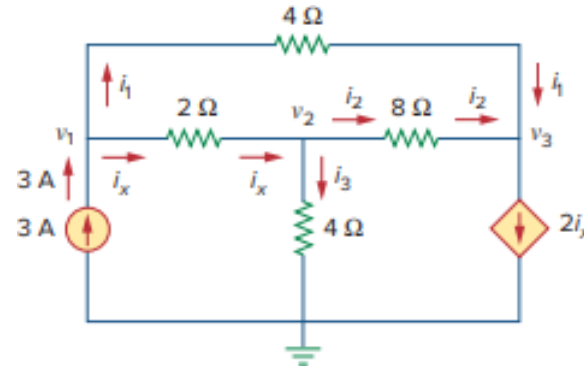
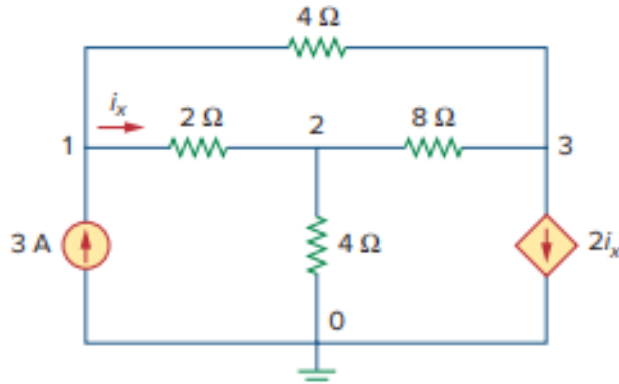
$$8 \left[\frac{v_1 - 5}{2} + \frac{v_1 - 0}{8} + \frac{v_1 + 3}{4} \right] = 0$$

$$\text{leads to } (4v_1 - 20) + (v_1) + (2v_1 + 6) = 0$$

$$7v_1 = +14, \quad v_1 = +2 \text{ V}, \quad i_{8\Omega} = \frac{v_1}{8} = \frac{2}{8} = \mathbf{0.25 \text{ A}}$$

Nodal Analysis

Determine the voltages at the nodes in the circuit.



At node 1, $3 = i_1 + i_x \Rightarrow 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$

$3v_1 - 2v_2 - v_3 = 12$ -----1

At node 2, $i_x = i_2 + i_3 \Rightarrow \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$

$-4v_1 + 7v_2 - v_3 = 0$ -----2

At node 3, $i_1 + i_2 = 2i_x \Rightarrow \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$

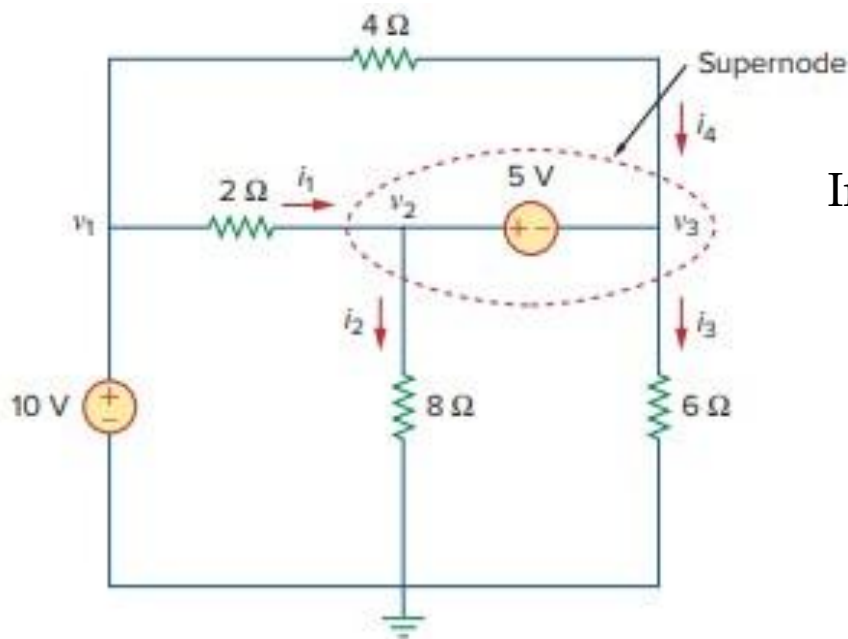
$2v_1 - 3v_2 + v_3 = 0$ -----3

Solve the 3 simultaneous equations:

$v_1 = 4.8 \text{ V}$

$v_2 = 2.4 \text{ V}$

$v_3 = -2.4 \text{ V}$



In this circuit nodes 2 and 3 form a supernode.

Case 1: A voltage source is connected between the reference node and a non reference node.

$$V_1 = 10 \text{ V} \quad \text{-----} 1$$

Case 2: A voltage source (dependent or independent) is connected between two non reference nodes, the two non reference nodes form a generalized node or supernode; we apply both KCL and KVL to determine the node voltages.

Super Node

An essential component of nodal analysis is applying KCL, which requires knowing the current through each element.

There is no way of knowing the current through a voltage source in advance.

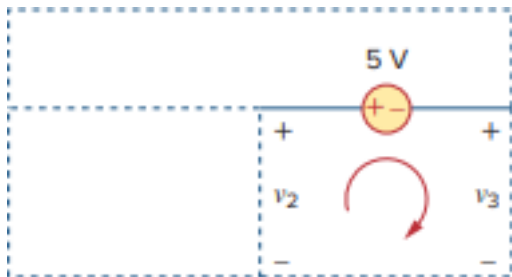
However, KCL must be satisfied at a supernode like any other node.

Hence, at the supernode, $i_1 + i_4 = i_2 + i_3$

or

$$\frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = \frac{v_2 - 0}{8} + \frac{v_3 - 0}{6} \quad \text{-----}2$$

To apply KVL to the supernode, redraw the circuit as shown and write the equation using KVL.



$$-v_2 + 5 + v_3 = 0 \quad \Rightarrow \quad v_2 - v_3 = 5 \quad \text{-----}3$$

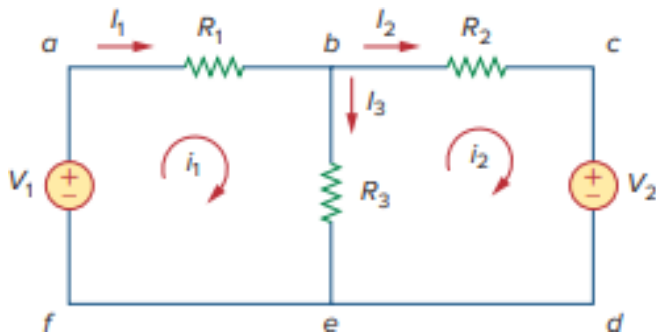
Solving these 3 equations, you can find the node voltages.

Mesh analysis

Mesh analysis applies KVL to find unknown currents.

Steps to determine Node Voltages:

1. Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes.
3. Use Ohm's law to express the voltages in terms of the mesh currents.
4. Solve the resulting n simultaneous equations to get the mesh currents.



Mesh 1,

$$-V_1 + R_1 i_1 + R_3 (i_1 - i_2) = 0$$

$$(R_1 + R_3) i_1 - R_3 i_2 = V_1 \quad \text{-----1}$$

Mesh 2,

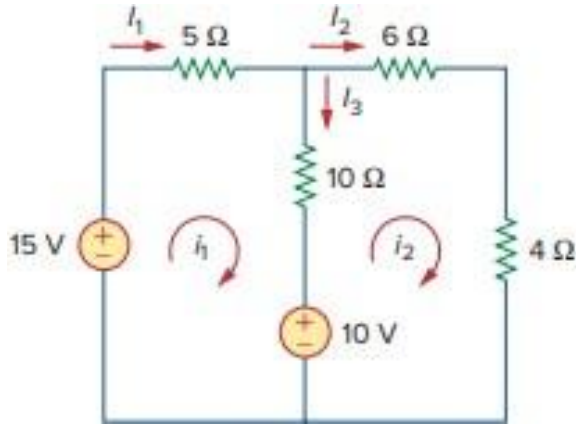
$$R_2 i_2 + V_2 + R_3 (i_2 - i_1) = 0$$

$$-R_3 i_1 + (R_2 + R_3) i_2 = -V_2 \quad \text{-----2}$$

Solve the 2 simultaneous equations to get i_1 and i_2 .

Mesh analysis

In the circuit below, find the branch currents I_1 , I_2 , and I_3 using mesh analysis.



Mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

Mesh 2,

$$3i_1 - 2i_2 = 1 \quad \text{-----1}$$

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

$$i_1 = 2i_2 - 1 \quad \text{-----2}$$

Solve the 2 simultaneous equations:

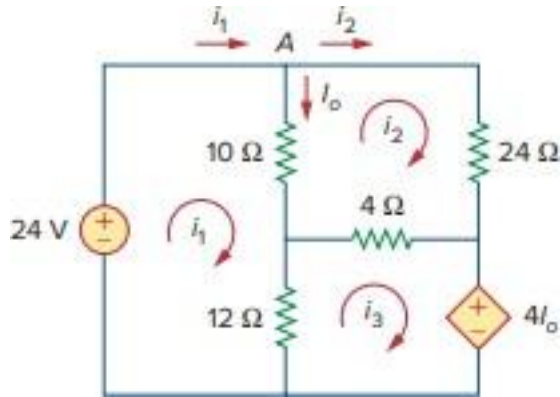
$$I_1 = 1\text{A}$$

$$I_2 = 1\text{A}$$

Hence $I_3 = 0$

Mesh analysis

Use mesh analysis to find the current I_o in the circuit .



Mesh 3,

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

At node A, $i_1 - i_2 = I_o$

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

Mesh 1,

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

$$-i_1 - i_2 + 2i_3 = 0 \quad \text{-----3}$$

Mesh 2,

$$11i_1 - 5i_2 - 6i_3 = 12 \quad \text{-----1}$$

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$$

$$-5i_1 + 19i_2 - 2i_3 = 0 \quad \text{-----2}$$

Solve the 3 simultaneous equations:

$$i_1 = 2.25\text{A}$$

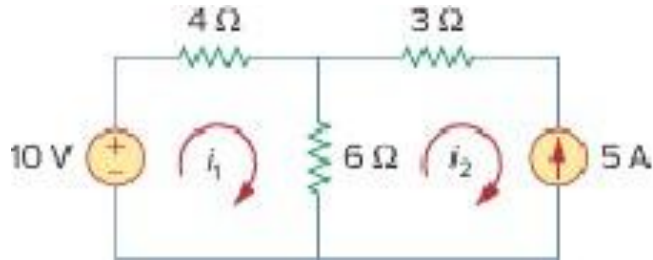
$$i_2 = 0.75\text{A}$$

$$I_3 = 1.5\text{A}$$

Hence $I_o = 1.5\text{A}$

Mesh analysis

Find the loop currents.



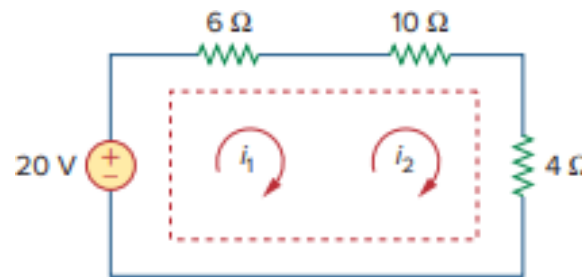
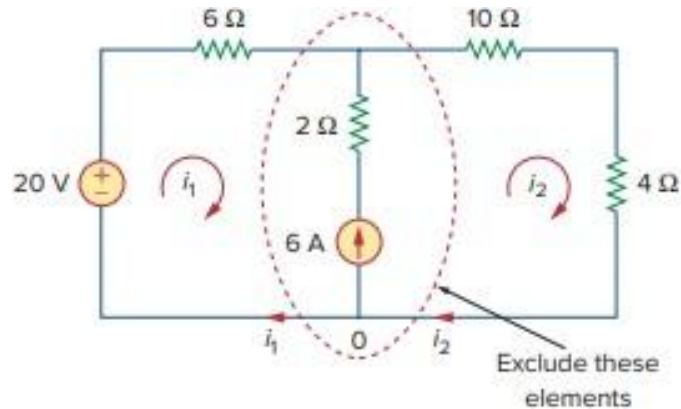
$$i_2 = -5\text{ A}$$

$$-10 + 4i_1 + 6(i_1 - i_2) = 0 \quad \Rightarrow \quad i_1 = -2\text{ A}$$

Supermesh

A supermesh results when two meshes have a (dependent or independent) **current source in common**.

A supermesh requires the application of both KVL and KCL.



We create a supermesh and apply KVL to the supermesh.

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0 \quad \text{or} \quad 6i_1 + 14i_2 = 20 \quad \text{-----}1$$

Apply KCL to a node in the branch where the two meshes intersect. Applying KCL to node 0,

$$i_2 = i_1 + 6 \quad \text{-----}2$$

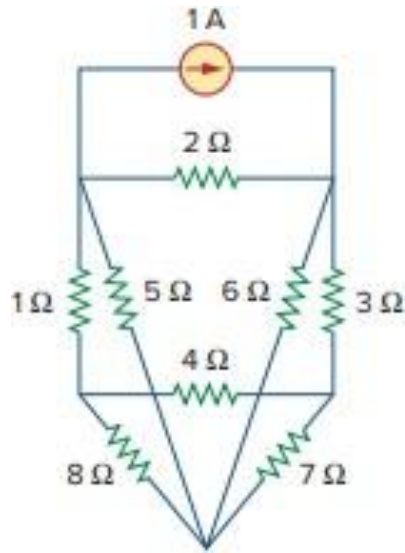
Solving equations 1 & 2 we get, $i_1 = -3.2 \text{ A}$, $i_2 = 2.8 \text{ A}$

Mesh analysis

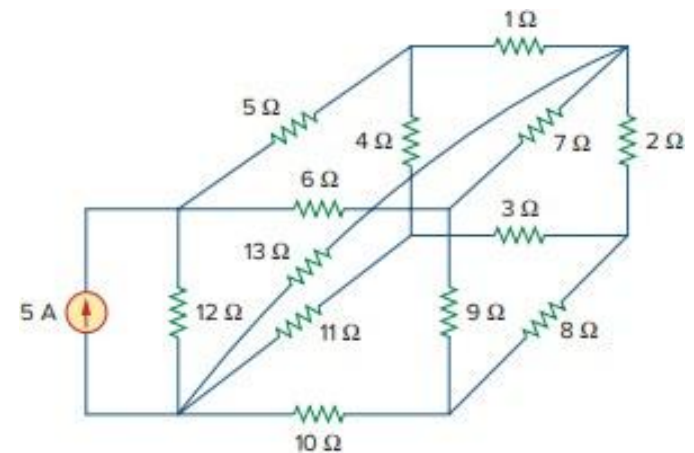
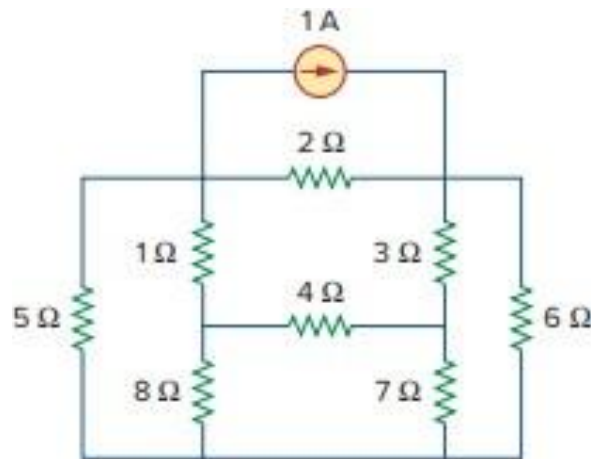
Mesh analysis is applicable to a circuit that is planar.

A planar circuit is one that can be drawn in a plane with no branches crossing one another; otherwise it is non-planar.

A circuit may have crossing branches and still be planar if it can be redrawn such that it has no crossing branches.



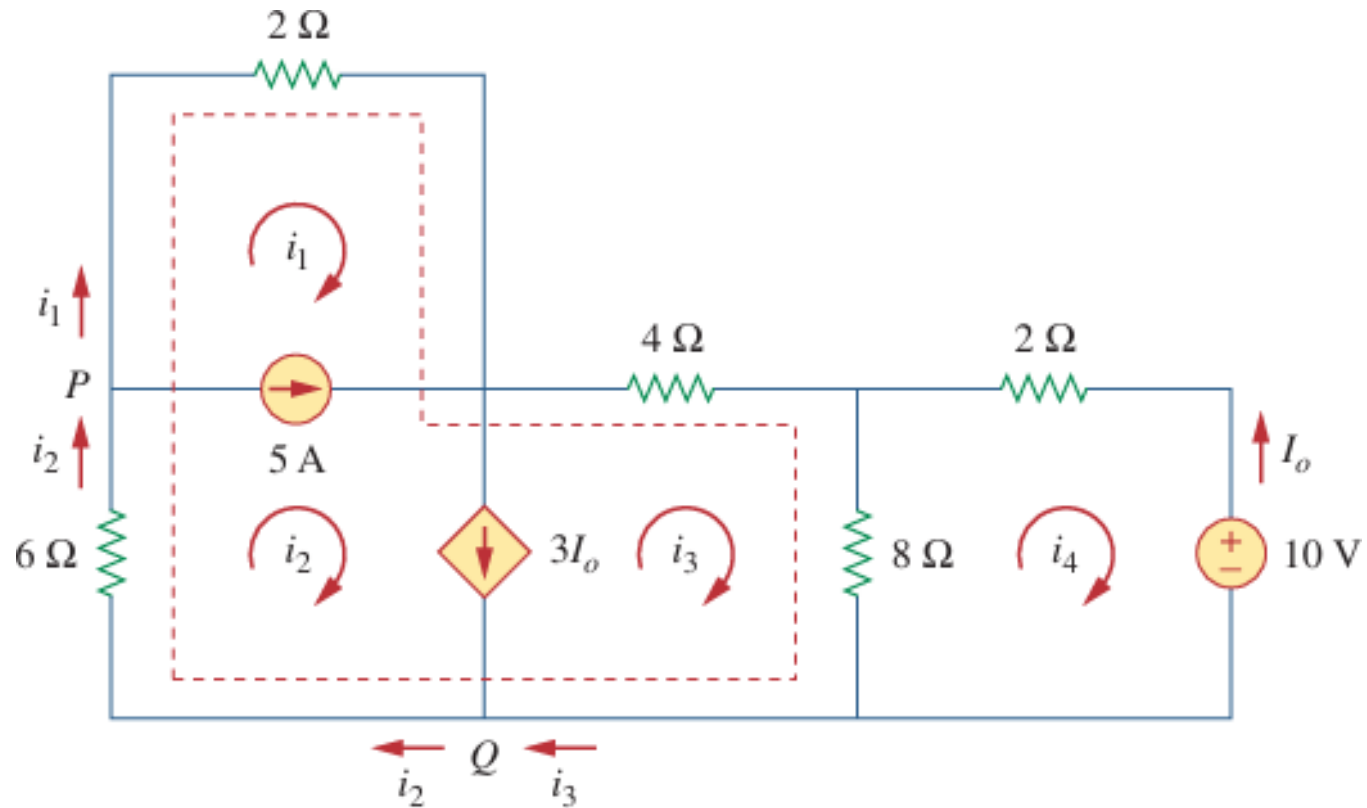
planar



non-planar

Supermesh

For the circuit below, calculate i_1 , i_2 , i_3 , i_4 .



Supermesh

Meshes 1 and 2 form a supermesh since they have an independent current source in common.

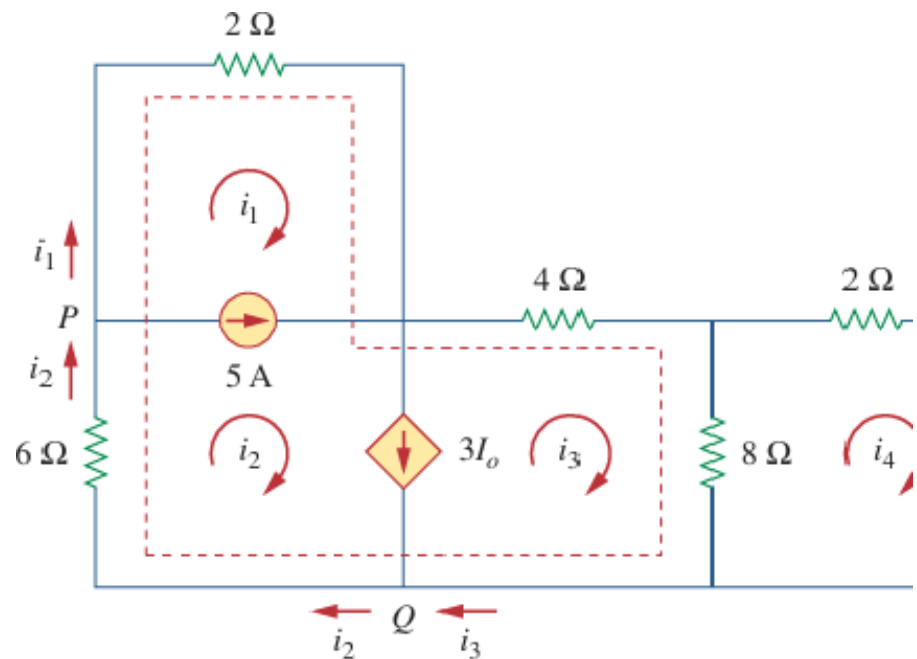
Also, meshes 2 and 3 form another supermesh because they have a dependent current source in common.

The two supermeshes intersect and form a larger supermesh.

Apply KVL to the larger supermesh,

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

$$i_1 + 3i_2 + 6i_3 - 4i_4 = 0 \quad \text{---1}$$



Supermesh

For the independent current source, we apply KCL to node P:

$$i_2 = i_1 + 5 \quad \text{-----} \quad 2$$

For the dependent current source, we apply KCL to node Q:

$$i_2 = i_3 + 3I_o$$

But $I_o = -i_4$, hence,

$$i_2 = i_3 - 3i_4 \quad \text{-----} \quad 3$$

Applying KVL in mesh 4,

$$2i_4 + 8(i_4 - i_3) + 10 = 0$$

$$5i_4 - 4i_3 = -5 \quad \text{-----} \quad 4$$

Solving the eqns 1, 2, 3, 4 we find

$$i_1 = -7.5 \text{ A}, \quad i_2 = -2.5 \text{ A}, \quad i_3 = 3.93 \text{ A}, \quad i_4 = 2.143 \text{ A}$$

Thank You